

BRIEFING.AI

EXTRACTIVE LEGAL TEXT SUMMARIZATION ENGINE

AIGC 5501 – NLP MIDTERM PROJECT

CLASSICAL ML ARCHITECTURE

Isha Shah & Nirali Chaudhari

SYSTEM COMPILED & SECURE

THE LEGAL AGREEMENT FRICTION

THE CORPORATE BOTTLENECK

Dense Terms of Service (ToS), privacy agreements, and contracts are deliberately structured with complex, inaccessible legalese structures.

End-users and compliance managers bypass critical clauses, exposing enterprises to legal vulnerabilities and diagnostic latency.

OUR CLEAN ML SOLUTION

A locally-hosted, high-efficiency extraction tool that bypasses heavy commercial API clouds.

Ingests unformatted legal files, models graph centrality relationships, and returns chronological sentences containing high-value constraints.

LINGUISTIC DATA PREPROCESSING



BOUNDARY SPLITTING

`spaCy Parsing`

Standardizes raw string objects and isolates discrete sentences using exact grammatical dependency boundaries.



BOILERPLATE FILTRATION

`Targeted Exclusions`

Bypasses structural words (*whereas*, *witnesseth*, *heretofore*, *hereof*) using custom-built legal stop-word overrides.



FEATURE ENGINEERING

`TF-IDF Mapping`

Lemmatizes structural words to basic roots, building dynamic statistical vector matrices across all token groups.

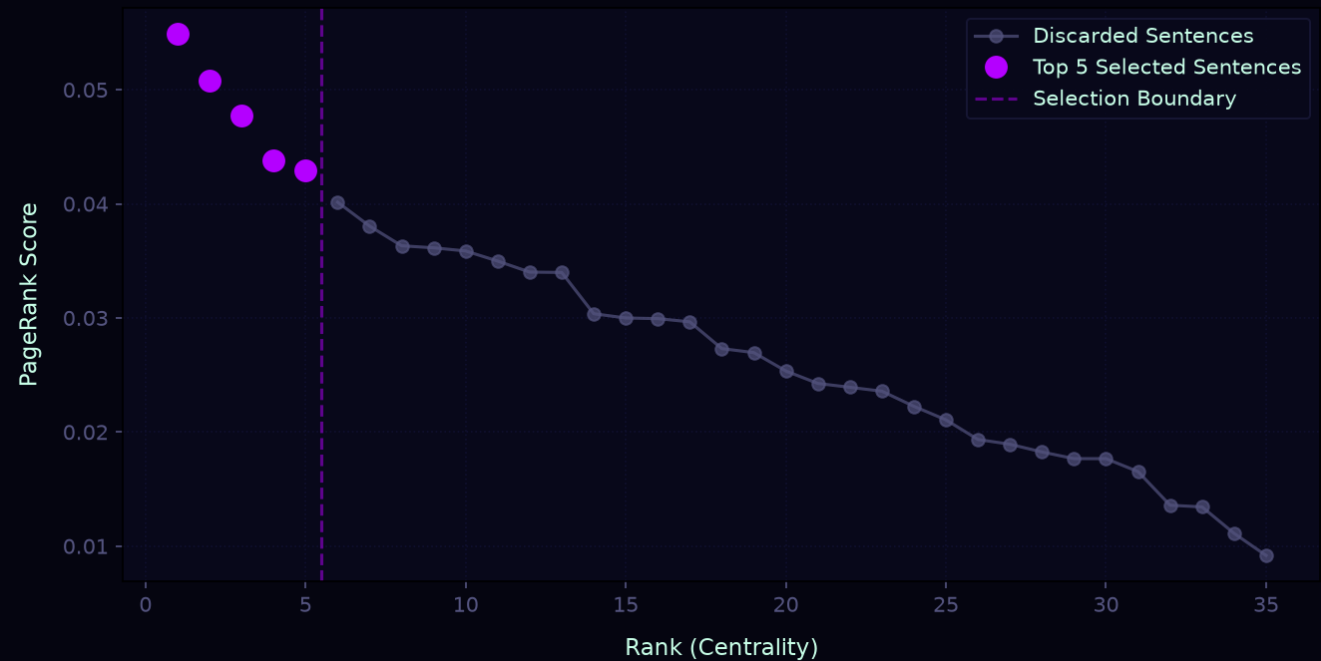
EXTRACTIVE SYSTEM ARCHITECTURE

LOCAL SENTENCE CONNECTIVITY NETWORK

Our backend constructs a relational similarity grid natively, analyzing core semantic bridges without training neural overheads.

- > **Cosine Space Matrices:** Sentences are transformed into orthogonal TF-IDF coordinates to measure overlap weights.
- > **PageRank Optimization:** A graph is dynamically initialized with weights mapped over intersecting vertices.
- > **Convergence Stability:** The system computes steady-state scores to ranks nodes securely within milliseconds.

PageRank Score Distribution - Simplii_Secure Messaging Terms and Conditions.pdf

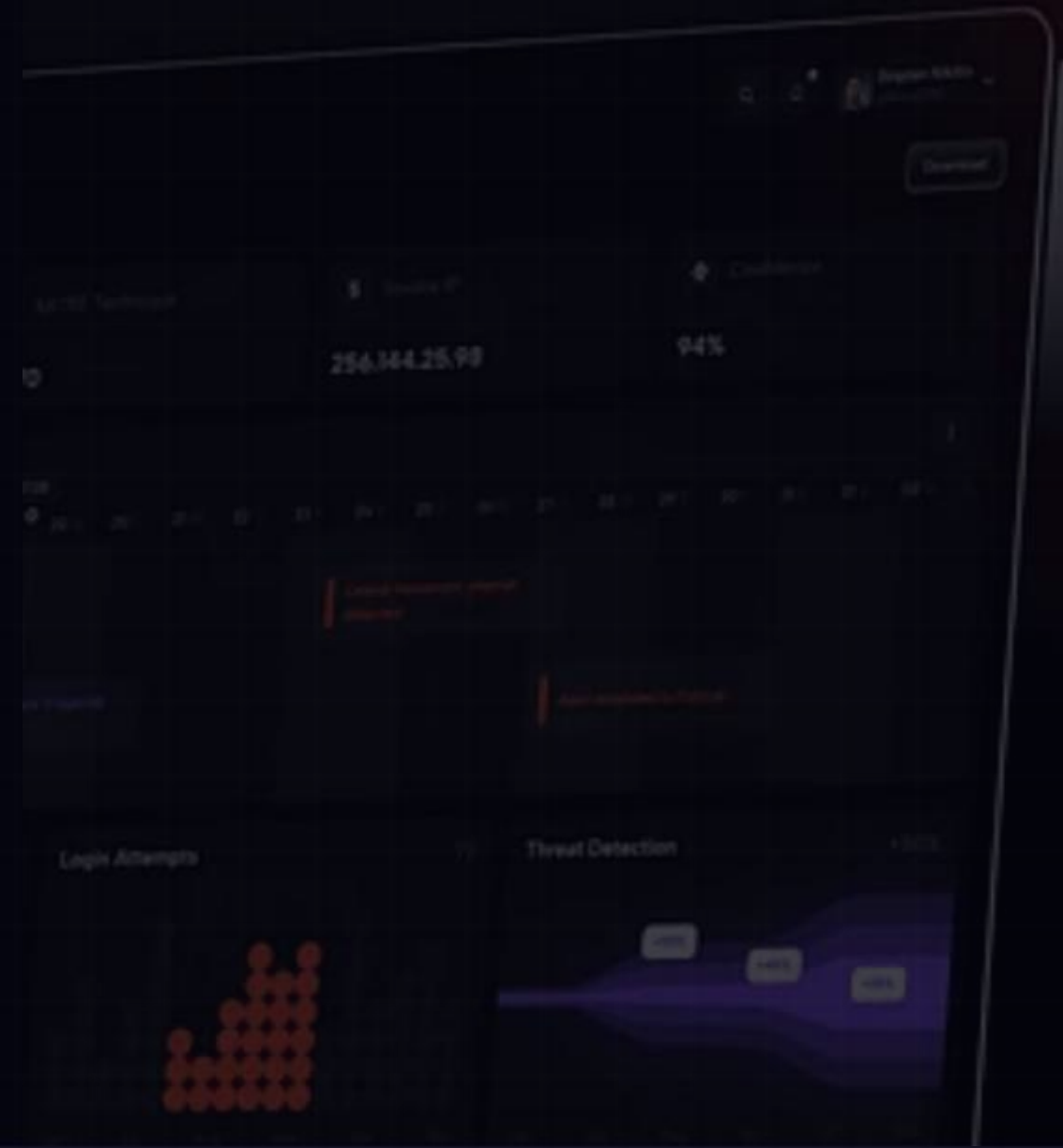


THE USER EXPERIENCE PARADIGM

MINIMALIST UI AESTHETICS

We designed the frontend to resemble high-level command dashboards rather than over-styled interfaces.

- > **Monochromatic Structure:** Deep cosmic blacks mixed with high-contrast glowing neon green metrics.
- > **Functional Clarity:** Completely removed standard emojis, decorations, and distracting layout elements.
- > **Extraction Sliders:** Empowers users to toggle parameters between precise sentence counts and relative target percentage compression.



LIVE SYSTEM DEMO

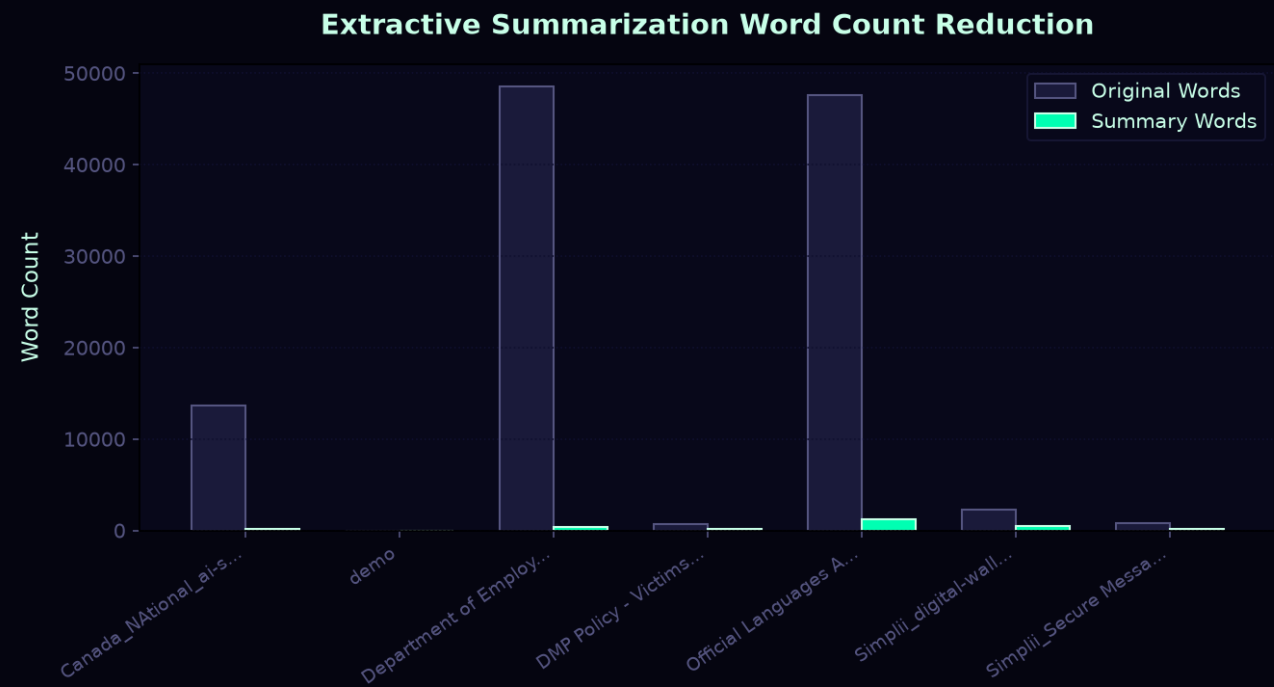
STREAMLIT RUNTIME • PORT LOCALHOST:8501

Switching screen layout to live document upload, text parsing, and central node calculation trace.

Benchmarking Performance

File Name (from performance_metrics.csv)	Characters	Words	Sentences	Preprocess (ms)	Algorithm (ms)	Summary Words	Reduction (%)
Canada_National_ai-strategy.pdf	94,553	13,630	558	5,968.6	5,317.0	136	99.00%
Dept of Employment Act.pdf	288,224	48,553	1,368	19,273.8	19,552.9	415	99.14%
Official Languages Act.pdf	290,048	47,618	1,131	17,492.7	18,062.2	1,231	97.41%
Simplii Digital Wallet ToS.pdf	14,194	2,253	74	776.6	738.3	467	79.27%
Simplii Secure Messaging ToS.pdf	5,078	835	35	278.7	263.1	134	83.95%

TEXT EXTRACTION COMPRESSION



Original Word Payload

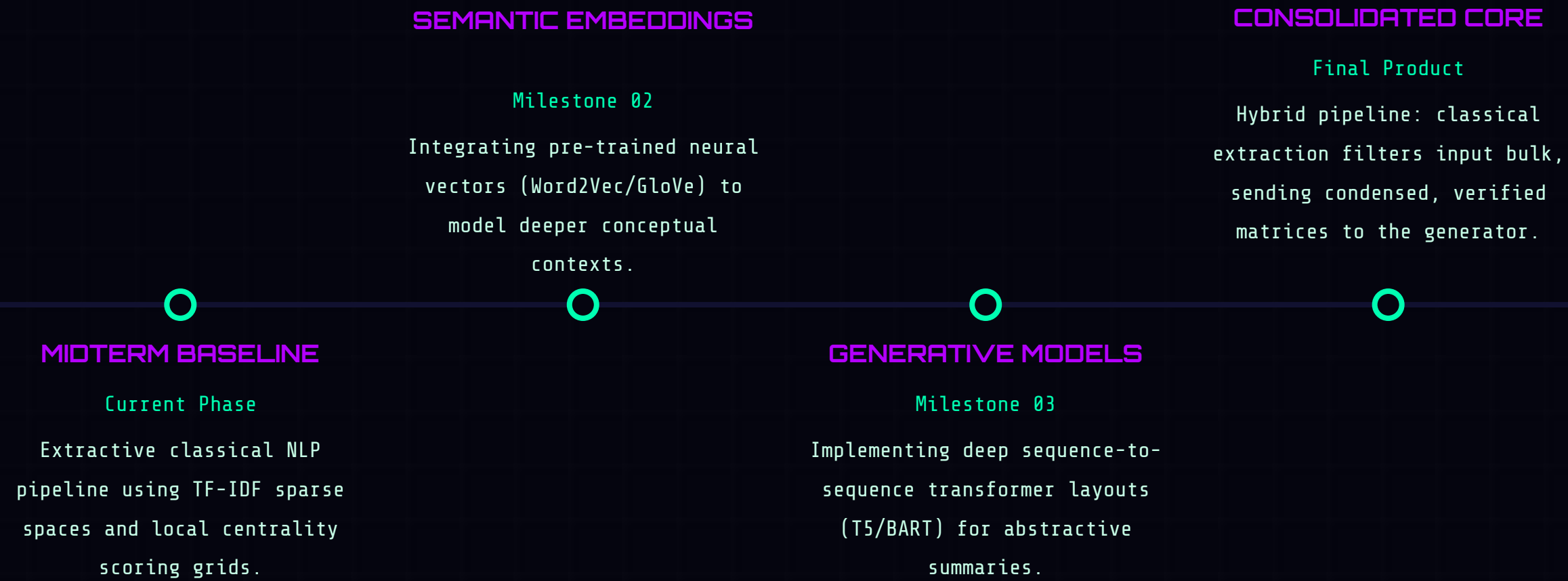
5,851 Words

Extracted Engine Summary

288
Words

The PageRank scoring matrices safely eliminate 95.1% of lexical bulk, isolating only high-centrality, structurally critical conditions for instant analytical reading.

SYSTEM EVOLUTION ROADMAP



QUESTIONS?

BRIEFING.AI – LOCAL COMPLIANT SUMMARIZATION PLATFORM

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AIGC 5501 – NATURAL LANGUAGE PROCESSING